KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 3 REPORT

Data Cleaning

Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

Team Member Name	Roll Number
Anireetha M	25BAD009

1. Week 3 Contribution

Researched suitable transformation techniques for the dataset.

2. Work Carried Out

During Week 3, my individual contribution focused on researching suitable data-cleaning techniques for the UCI Drug Consumption dataset. I reviewed the missing-value patterns and the data types of the affected columns, then studied why different imputation methods were suitable for ordinal, continuous and categorical variables. I also examined methods for detecting potential outliers and considered why they should be investigated before deciding whether to remove them.

3. Techniques Researched and Selected

- Median imputation – Suitable for Age, Education and Nscore. Age and Education are ordinal-coded variables, while Nscore is a continuous personality score. The median is less affected by extreme values than the mean and helps preserve the typical value of each column.
- Mode imputation – Suitable for Gender because it is a binary categorical variable with no numerical order. Replacing missing entries with the most frequent category preserves a valid category rather than creating an artificial value.
- Boxplot method – Suitable for visually inspecting unusual values in the seven personality-score columns: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS. Values beyond the boxplot whiskers can be flagged for further checking.
- Z-score method – Suitable for identifying values that are far from the mean in the selected numerical columns. The notebook used $|Z| > 3$ as the potential-outlier threshold.
- Outlier handling decision – Potential outliers were reported rather than automatically deleted, because extreme personality scores may represent genuine responses and not data-entry errors.

4. Outcome of My Contribution

The notebook applied median imputation to Age, Education and Nscore, and mode imputation to Gender. The missing-value check then showed zero remaining missing values. Boxplots and Z-scores identified potential outliers: Nscore (4), Escore (6), Oscore (2), Ascore (5), Cscore (6), Impulsive (7) and SS (0), using $|Z| > 3$. These observations were retained for further investigation. The final cleaned dataset contained 1,885 rows and 32 columns, with zero missing values and zero duplicate rows. This report focuses on researching and explaining the suitable cleaning techniques reflected in the notebook.